

SYNAPSE AI Agent Specification

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Chapter 1

Introduction

This document provides a comprehensive technical specification for the AI Agent layer in SYNAPSE. The SYNAPSE platform integrates an advanced Agentic AI system built on LangChain ReAct agents, enabling autonomous execution of complex multi-step tasks. This specification covers architecture, agent types, tool definitions, memory systems, safety mechanisms, and monitoring capabilities.

The AI Agent layer is the intelligent core of SYNAPSE, enabling users to delegate complex workflows to autonomous AI systems that can reason, act, observe, and iterate toward task completion. All agents operate within defined safety boundaries with comprehensive logging and cost tracking.

Chapter 2

Agent Architecture Overview

2.1 ReAct Pattern Foundation

The SYNAPSE AI Agent layer is built on the ReAct (Reasoning + Acting) pattern, a powerful paradigm that combines large language models' reasoning capabilities with tool-based action execution. The ReAct pattern operates through a structured loop that maximizes transparency and controllability.

The core loop consists of four sequential stages:

1. **Thought:** The agent reasons about the current state, available tools, and optimal next steps. This represents internal cognition where the agent formulates a plan.
2. **Action:** Based on its reasoning, the agent selects a specific tool and constructs appropriate input parameters.
3. **Observation:** The tool executes and returns a result, which the agent receives as structured feedback.
4. **Repeat:** The agent evaluates the observation against the task goal and either continues the loop or terminates with a final answer.

This transparent loop allows developers to inspect agent decision-making at each step, improving debugging and safety oversight.

2.2 Agent Loop Mechanics

The agent loop in SYNAPSE operates as follows:

```
1 while not task_complete:
2     thought = agent.reason(context, goal)
3     action, tool_name, tool_input = agent.select_action(thought)
4     observation = tools[tool_name].execute(tool_input)
5     context = agent.update_context(observation)
6     task_complete = agent.is_goal_reached(observation)
```

Each iteration is atomic and traceable. The agent maintains execution context including:

- Current task objective and subtasks
- Conversation history and memory
- Tool availability and constraints
- Token budget and cost limits
- Timeout tracking per tool and overall agent run

2.3 LangChain AgentExecutor Configuration

The LangChain AgentExecutor manages the agent lifecycle and coordinates tool execution. Key configuration parameters include:

```
1 from langchain.agents import AgentExecutor, create_react_agent
2 from langchain_openai import ChatOpenAI
3 from langchain.prompts import PromptTemplate
4
5 executor = AgentExecutor.from_agent_and_tools(
6     agent=agent,
7     tools=tool_list,
8     verbose=True,
9     max_iterations=10,
10    max_execution_time=300,
11    early_stopping_method="force",
12    return_intermediate_steps=True,
13    handle_parsing_errors=True
14 )
```

Critical AgentExecutor parameters:

- **max_iterations:** Maximum ReAct loop cycles (default 10 per SYNAPSE spec)
- **max_execution_time:** Hard timeout in seconds (300s per agent run)
- **early_stopping_method:** Handles max iterations or timeout (force or generate)
- **return_intermediate_steps:** Captures all thought-action-observation records
- **handle_parsing_errors:** Graceful error recovery when agent output is malformed

2.4 Memory Management

Agent memory is essential for maintaining context across multiple interactions and enabling long-running task execution. SYNAPSE implements multiple memory strategies:

- **Execution Memory:** Current task context and intermediate results within single agent run
- **Conversation Memory:** User interaction history across multiple agent invocations
- **Semantic Memory:** Vector-based retrieval for relevant knowledge and past solutions
- **Entity Memory:** Tracking of mentioned technologies, projects, and domain concepts

Memory state is serialized and stored in PostgreSQL for persistence and retrieval across sessions.

Chapter 3

Agent Types

SYNAPSE defines five specialized agent types, each optimized for specific task categories:

3.1 ResearchAgent

The ResearchAgent autonomously searches and synthesizes information from multiple sources to answer complex research queries.

Capabilities:

- Search knowledge base with semantic filtering
- Fetch and summarize academic articles
- Cross-reference information across sources
- Generate comprehensive research summaries
- Track source citations for verification

Primary Tools: `search_knowledge_base`, `fetch_articles`, vector store retrieval

Task Examples:

- Analyze technology landscape for a given domain
- Compile literature review on emerging trends
- Compare competing solutions and frameworks

3.2 DocumentAgent

The DocumentAgent generates professional documents in multiple formats (PDF, PowerPoint, Word) based on structured content specifications.

Capabilities:

- Generate styled PDF reports with charts and tables
- Create PowerPoint presentations with layouts and images
- Produce Word documents with TOC and formatting
- Generate Markdown documentation
- Apply consistent branding and styling

Primary Tools: generate_pdf, generate_ppt, generate_word_doc, markdown_generation

Task Examples:

- Create quarterly technology trend report
- Generate project proposal document
- Produce training materials and documentation

3.3 TrendAgent

The TrendAgent analyzes technology trends, tracks adoption patterns, and identifies emerging areas of significance.

Capabilities:

- Analyze technology trend data over time periods
- Fetch trending repositories and projects from GitHub
- Retrieve and analyze academic papers from arXiv
- Synthesize trend information into insights
- Generate trend reports and forecasts

Primary Tools: analyze_trends, search_github, fetch_arxiv_papers, get_technology_trend

Task Examples:

- Identify emerging frameworks in a programming domain
- Track adoption of new technologies
- Analyze GitHub repository popularity trends

3.4 ProjectAgent

The ProjectAgent scaffolds new software projects with complete directory structures, boilerplate code, and configuration files.

Capabilities:

- Generate project structures for multiple frameworks
- Create boilerplate code and configuration
- Set up dependency management and build systems
- Initialize version control and documentation
- Generate README and setup instructions

Primary Tools: create_project, file_system_operations

Task Examples:

- Bootstrap FastAPI microservice with middleware
- Create Next.js application with routing and layout
- Generate Django REST API with models and endpoints

3.5 SchedulerAgent

The SchedulerAgent manages automation workflows, scheduling, and task orchestration across the SYNAPSE platform.

Capabilities:

- Schedule recurring agent runs and reports
- Manage dependencies between tasks
- Monitor and retry failed automation workflows
- Trigger agents based on external events or schedules
- Aggregate results from multiple agent executions

Primary Tools: Celery task scheduling, workflow orchestration

Task Examples:

- Generate weekly trend report every Monday
- Run research queries on schedule with result email delivery
- Chain multiple agents (ResearchAgent then DocumentAgent)

Chapter 4

Tool Definitions

SYNAPSE agents interact with external systems and data sources through a standardized tool interface. Each tool is defined by name, description, input/output schemas, and implementation details.

4.1 Knowledge Base Tools

4.1.1 search_knowledge_base

Search the SYNAPSE knowledge base using semantic similarity and keyword matching.

Tool Definition:

```
1 class SearchKnowledgeBaseInput(BaseModel):
2     query: str = Field(..., description="Search_query")
3     limit: int = Field(default=10, description="Max_results")
4     filters: Optional[Dict[str, Any]] = Field(
5         default=None,
6         description="Optional_filters_(source,_date_range,_category)"
7     )
8
9 class SearchResult(BaseModel):
10     id: str
11     title: str
12     content: str
13     relevance_score: float
14     source: str
15     metadata: Dict[str, Any]
```

Implementation Notes:

- Uses pgvector with PostgreSQL for semantic search
- Supports BM25 keyword matching as fallback
- Filters by source (internal docs, articles, code)
- Returns ranked results with relevance scores
- Caches results for 1 hour to reduce latency

4.1.2 fetch_articles

Retrieve articles from external sources (arXiv, Medium, etc.) within a date range.

Tool Definition:

```

1 class FetchArticlesInput(BaseModel):
2     topic: str = Field(..., description="Article topic")
3     date_range: Tuple[date, date] = Field(
4         ..., description="Start and end dates"
5     )
6     limit: int = Field(default=20, description="Max articles to fetch")
7     sources: Optional[List[str]] = Field(
8         default=["arxiv", "medium"],
9         description="Article sources"
10    )
11
12 class ArticleMetadata(BaseModel):
13     url: str
14     title: str
15     authors: List[str]
16     published_date: datetime
17     summary: str
18     source: str

```

Implementation Notes:

- Integrates with arXiv, Medium, and Hacker News APIs
- Caches fetched articles in PostgreSQL
- Deduplicates results across sources
- Timeout: 30 seconds per API call

4.2 Trend Analysis Tools

4.2.1 analyze_trends

Analyze technology adoption and trend patterns over specified time periods.

Tool Definition:

```

1 class AnalyzeTrendsInput(BaseModel):
2     technologies: List[str] = Field(
3         ..., description="Technology names to analyze"
4     )
5     period: str = Field(
6         default="12m",
7         description="Time period (1m, 3m, 6m, 12m, 5y)"
8     )
9     metrics: Optional[List[str]] = Field(
10        default=["search_volume", "github_stars", "job_postings"],
11        description="Metrics to analyze"
12    )
13
14 class TrendAnalysis(BaseModel):
15     technology: str
16     period: str
17     trend_direction: str
18     growth_rate: float
19     metrics_timeline: List[Dict[str, Any]]
20     insights: str

```

Implementation Notes:

- Aggregates data from multiple sources (Google Trends, GitHub, Job APIs)
- Calculates growth rates and trend direction
- Generates insights using statistical analysis
- Caches results for 24 hours

4.2.2 get_technology_trend

Retrieve current trend information for a specific technology.

Tool Definition:

```

1 class GetTechnologyTrendInput(BaseModel):
2     tech_name: str = Field(..., description="Technology_name")
3     period: str = Field(default="12m", description="Analysis_period")
4
5 class TrendData(BaseModel):
6     name: str
7     trend_score: float
8     growth_percentage: float
9     popularity_rank: int
10    last_updated: datetime
11    sources: List[str]
```

Implementation Notes:

- Real-time data from GitHub API and Google Trends
- Trend score combines multiple signals (0-100 scale)
- Updates cached data hourly

4.3 Document Generation Tools

4.3.1 generate_pdf

Create professional PDF documents with styled content, tables, and charts.

Tool Definition:

```

1 class GeneratePDFInput(BaseModel):
2     title: str = Field(..., description="Document_title")
3     sections: List[Dict[str, Any]] = Field(
4         ..., description="Document_sections"
5     )
6     content: str = Field(..., description="Main_content")
7     metadata: Optional[Dict[str, str]] = Field(
8         default=None, description="Author, version, etc."
9     )
10
11 class PDFOutput(BaseModel):
12     file_path: str
13     file_size: int
14     pages: int
15     url: str
```

Implementation Notes:

- Uses ReportLab 4.x for PDF generation

- Supports tables, charts, images, headers/footers
- Custom styling with fonts and colors
- Generates TOC automatically
- Output stored in PostgreSQL large object storage

4.3.2 generate_ppt

Create PowerPoint presentations with multiple slide layouts.

Tool Definition:

```
1 class GeneratePPTInput(BaseModel):
2     title: str = Field(..., description="Presentation_title")
3     slides: List[Dict[str, Any]] = Field(
4         ..., description="Slide_definitions"
5     )
6     theme: str = Field(default="default", description="Theme_name")
7
8 class PPTOutput(BaseModel):
9     file_path: str
10    slide_count: int
11    url: str
```

Implementation Notes:

- Uses python-pptx 0.6.x library
- Supports title, bullet, image, and chart slide layouts
- Built-in themes with consistent branding
- Image embedding and sizing

4.3.3 generate_word_doc

Create formatted Word documents with styles and structure.

Tool Definition:

```
1 class GenerateWordDocInput(BaseModel):
2     title: str = Field(..., description="Document_title")
3     content: str = Field(..., description="Document_content")
4     toc: bool = Field(default=True, description="Include_TOC")
5
6 class WordDocOutput(BaseModel):
7     file_path: str
8     word_count: int
9     url: str
```

Implementation Notes:

- Uses python-docx 0.8.x library
- Auto-generates table of contents
- Applies heading hierarchy and styles
- Supports embedded images and tables

4.4 Project Creation Tools

4.4.1 create_project

Bootstrap new software projects with complete structure and boilerplate.

Tool Definition:

```

1 class CreateProjectInput(BaseModel):
2     project_type: str = Field(
3         ...,
4         description="Project_type_(django_rest,fastapi,nextjs,react_lib,datascience)"
5     )
6     name: str = Field(..., description="Project_name")
7     features: List[str] = Field(
8         default=[],
9         description="Additional_features_(auth,testing,ci_cd)"
10    )
11    destination_path: Optional[str] = Field(
12        default=None, description="Where_to_create_project"
13    )
14
15 class ProjectOutput(BaseModel):
16     project_path: str
17     files_created: int
18     structure: Dict[str, Any]
19     next_steps: List[str]

```

Implementation Notes:

- Supports: Django REST API, FastAPI, Next.js, React component library, Python data science
- Generates complete directory structure
- Creates boilerplate code and configuration
- Includes README with setup instructions
- Initializes git repository and .gitignore

4.5 Cloud Storage Tools

4.5.1 upload_to_drive

Upload files to Google Drive with folder organization.

Tool Definition:

```

1 class UploadToDriveInput(BaseModel):
2     file_path: str = Field(..., description="Local_file_path")
3     folder: str = Field(default="SYNAPSE", description="Drive_folder_path")
4     public: bool = Field(default=False, description="Make_file_public")
5
6 class DriveUploadOutput(BaseModel):
7     file_id: str
8     file_url: str
9     folder_id: str

```

Implementation Notes:

- Uses Google Drive API v3
- Creates folder hierarchy automatically

- Timeout: 60 seconds per upload
- Supports files up to 5GB

4.5.2 upload_to_s3

Upload files to Amazon S3 with configurable storage classes.

Tool Definition:

```

1 class UploadToS3Input(BaseModel):
2     file_path: str = Field(..., description="Local_file_path")
3     bucket: str = Field(..., description="S3_bucket_name")
4     key: str = Field(..., description="S3_object_key/path")
5     storage_class: str = Field(
6         default="STANDARD",
7         description="S3_storage_class"
8     )
9
10 class S3UploadOutput(BaseModel):
11     bucket: str
12     key: str
13     object_url: str
14     etag: str

```

Implementation Notes:

- Uses boto3 library
- Supports multipart upload for large files
- Configurable ACL and metadata
- Timeout: 120 seconds per upload

4.6 Communication Tools

4.6.1 send_email

Send emails via configured SMTP or service provider.

Tool Definition:

```

1 class SendEmailInput(BaseModel):
2     to: Union[str, List[str]] = Field(
3         ..., description="Recipient(s)"
4     )
5     subject: str = Field(..., description="Email_subject")
6     body: str = Field(..., description="Email_body_(HTML_or_plain_text)")
7     attachments: Optional[List[str]] = Field(
8         default=None, description="File_paths_to_attach"
9     )
10
11 class EmailOutput(BaseModel):
12     message_id: str
13     status: str
14     sent_at: datetime

```

Implementation Notes:

- Supports SMTP and SendGrid API

- Rate limit: 100 emails per hour per user
- Timeout: 30 seconds
- Attachment size limit: 25MB

4.7 Code Search Tools

4.7.1 search_github

Search GitHub repositories with language and quality filters.

Tool Definition:

```

1 class SearchGitHubInput(BaseModel):
2     query: str = Field(..., description="Search_query")
3     language: Optional[str] = Field(
4         default=None, description="Programming_language_filter"
5     )
6     stars_min: int = Field(
7         default=0, description="Minimum_star_count"
8     )
9     limit: int = Field(default=30, description="Result_limit")
10
11 class GitHubResult(BaseModel):
12     name: str
13     url: str
14     description: str
15     stars: int
16     language: str
17     last_updated: datetime

```

Implementation Notes:

- Uses GitHub REST API v3
- Sorts results by stars by default
- API rate limit: 60 requests per hour (higher with authentication)
- Caches results for 24 hours

4.7.2 fetch_arxiv_papers

Retrieve academic papers from arXiv with filtering.

Tool Definition:

```

1 class FetchArXivPapersInput(BaseModel):
2     query: str = Field(..., description="Search_query")
3     max_results: int = Field(default=50, description="Max_papers")
4     sort_by: str = Field(
5         default="relevance",
6         description="Sort_key_(relevance,submission_date)"
7     )
8
9 class ArXivPaper(BaseModel):
10     arxiv_id: str
11     title: str
12     authors: List[str]
13     published: datetime
14     summary: str

```

```
15 pdf_url: str
16 categories: List[str]
```

Implementation Notes:

- Uses arXiv public API (no authentication needed)
- Timeout: 30 seconds
- Caches results for 7 days
- No rate limiting (community guidelines apply)

Chapter 5

LangChain Integration

5.1 AgentExecutor Configuration

The LangChain AgentExecutor is the orchestration engine for SYNAPSE agents. Configuration involves several key components working in concert.

5.1.1 Tool Registration

Tools must be registered with the AgentExecutor in a standardized format:

```
1 from langchain.tools import Tool, StructuredTool
2
3 tools = [
4     StructuredTool.from_function(
5         func=search_knowledge_base,
6         name="search_knowledge_base",
7         description="Search internal knowledge base",
8         args_schema=SearchKnowledgeBaseInput,
9     ),
10    StructuredTool.from_function(
11        func=generate_pdf,
12        name="generate_pdf",
13        description="Generate PDF documents",
14        args_schema=GeneratePDFInput,
15    ),
16 ]
17
18 executor = AgentExecutor.from_agent_and_tools(
19     agent=agent,
20     tools=tools,
21     verbose=True,
22     max_iterations=10,
23     max_execution_time=300,
24     early_stopping_method="force",
25     return_intermediate_steps=True,
26     handle_parsing_errors=True,
27     trim_intermediate_steps=5
28 )
```

5.2 Prompt Templates

SYNAPSE uses custom prompt templates to guide agent reasoning and ensure consistent behavior.

5.2.1 System Prompt

The system prompt establishes agent role, capabilities, and constraints:

```

1 SYSTEM_PROMPT = """You are an AI agent in the SYNAPSE platform,
2 designed to autonomously execute complex multi-step tasks.
3 You have access to specialized tools for research, document
4 generation, trend analysis, and project scaffolding.
5
6 Guidelines:
7 1. Break complex tasks into logical subtasks
8 2. Use available tools to gather information
9 3. Cite sources for all information retrieved
10 4. Always validate data before using it
11 5. Respect user privacy and data security
12 6. Do not make assumptions without verification
13 7. Ask for clarification if task is ambiguous
14
15 Available tools: {tools}
16
17 Format your response as JSON with 'thought', 'action',
18 'action_input', and 'observation' fields."""

```

5.2.2 Human Prompt

The human prompt structure provides context for each agent invocation:

```

1 HUMAN_PROMPT = """Task: {task}
2
3 Current Date: {current_date}
4 User: {user_id}
5 Context: {context}
6
7 Previous steps taken:
8 {previous_steps}
9
10 Next steps to consider:
11 {next_steps_hint}"""

```

5.3 Output Parsers

Output parsing converts agent text responses into structured data:

```

1 from langchain.agents.output_parsers import ReActJsonSingleInputOutputParser
2
3 parser = ReActJsonSingleInputOutputParser()
4
5 parsed_output = parser.parse(agent_response)
6
7 class ParsedOutput(BaseModel):
8     thought: str
9     action: str
10     action_input: Dict[str, Any]
11     observation: Optional[str]
12     final_answer: Optional[str]

```

5.4 Streaming and Callbacks

SYNAPSE supports real-time streaming of agent thoughts and actions via callbacks:

```

1 from langchain.callbacks.streaming_stdout import StreamingStdOutCallbackHandler
2 from langchain.callbacks import StreamingStdOutCallbackHandler
3
4 class CustomAgentCallback(StreamingStdOutCallbackHandler):
5     def on_agent_action(self, action, **kwargs):
6         print(f"Agent Action: {action.tool}")
7         print(f"Input: {action.tool_input}")
8
9     def on_agent_finish(self, finish, **kwargs):
10        print(f"Final Answer: {finish.output}")
11
12 executor = AgentExecutor(
13     agent=agent,
14     tools=tools,
15     callbacks=[CustomAgentCallback()],
16 )

```

5.5 Token Counting and Cost Tracking

Token usage and cost tracking are integrated at the executor level:

```

1 from tiktoken import encoding_for_model
2
3 class CostTracker:
4     def __init__(self, model="gpt-4"):
5         self.model = model
6         self.encoder = encoding_for_model("gpt-4")
7         self.input_cost_per_1k = 0.03
8         self.output_cost_per_1k = 0.06
9
10    def estimate_tokens(self, text):
11        return len(self.encoder.encode(text))
12
13    def calculate_cost(self, input_tokens, output_tokens):
14        input_cost = (input_tokens / 1000) * self.input_cost_per_1k
15        output_cost = (output_tokens / 1000) * self.output_cost_per_1k
16        return input_cost + output_cost
17
18 tracker = CostTracker()
19 result = executor.invoke({"input": task})
20 total_cost = tracker.calculate_cost(
21     result["input_tokens"],
22     result["output_tokens"]
23 )

```

Chapter 6

Memory Systems

SYNAPSE implements four complementary memory systems to maintain context across agent invocations.

6.1 ConversationBufferWindowMemory

Stores the last N turns of conversation for immediate context.

```
1 from langchain.memory import ConversationBufferWindowMemory
2
3 memory = ConversationBufferWindowMemory(
4     k=10,
5     memory_key="chat_history",
6     return_messages=True
7 )
8
9 memory.save_context(
10     {"input": user_query},
11     {"output": agent_response}
12 )
13
14 context = memory.load_memory_variables({})
```

Characteristics:

- Stores last 10 conversation turns
- Lightweight and fast retrieval
- Suitable for short-term context
- Stored in PostgreSQL with TTL of 30 days

6.2 ConversationSummaryMemory

Summarizes conversation history to preserve key information while reducing token usage.

```
1 from langchain.memory import ConversationSummaryMemory
2 from langchain_openai import ChatOpenAI
3
4 memory = ConversationSummaryMemory(
5     llm=ChatOpenAI(model="gpt-3.5-turbo"),
6     buffer="",
7     memory_key="chat_summary"
8 )
```

```

9
10 memory.add_messages(messages)
11 summary = memory.load_memory_variables({})

```

Characteristics:

- Automatically summarizes conversation history
- Reduces context window usage by 70-80 percent
- Useful for long-running sessions
- Updated every 5 conversation turns

6.3 VectorStoreRetrieverMemory

Uses semantic similarity to retrieve relevant past interactions.

```

1 from langchain.memory import VectorStoreRetrieverMemory
2 from langchain_community.vectorstores import Chroma
3
4 vectorstore = Chroma(
5     embedding_function=embeddings,
6     collection_name="agent_memory"
7 )
8
9 memory = VectorStoreRetrieverMemory(
10     vectorstore=vectorstore,
11     memory_key="relevant_history",
12     input_key="input"
13 )
14
15 retrieved = memory.load_memory_variables({"input": user_query})

```

Characteristics:

- Semantic search across all past interactions
- Retrieves topN most relevant memories
- Powered by pgvector for efficiency
- Enables knowledge accumulation across sessions

6.4 Entity Memory

Tracks and updates named entities (technologies, projects, concepts) mentioned in interactions.

```

1 from langchain.memory import EntityMemory
2
3 entity_store = {
4     "technologies": {},
5     "projects": {},
6     "people": {},
7     "organizations": {}
8 }
9
10 def track_entity(entity_name, entity_type, properties):
11     if entity_name not in entity_store[entity_type]:

```

```
12     entity_store[entity_type][entity_name] = {  
13         "first_mention": datetime.now(),  
14         "mentions": 0,  
15         "properties": {}  
16     }  
17  
18     entity_store[entity_type][entity_name]["mentions"] += 1  
19     entity_store[entity_type][entity_name]["properties"].update(properties)
```

Characteristics:

- Maintains entity graphs for relationship mapping
- Tracks mention frequency and temporal patterns
- Enables context-aware agent decisions
- Stored as JSON in PostgreSQL

Chapter 7

RAG Pipeline

The Retrieval-Augmented Generation (RAG) pipeline enhances agent reasoning with grounded information from knowledge bases.

7.1 ConversationalRetrievalChain

The ConversationalRetrievalChain combines conversation history with document retrieval:

```
1 from langchain.chains import ConversationalRetrievalChain
2 from langchain_openai import ChatOpenAI
3 from langchain_community.vectorstores import Chroma
4
5 retriever = vectorstore.as_retriever(
6     search_type="similarity",
7     search_kwargs={"k": 5}
8 )
9
10 qa_chain = ConversationalRetrievalChain.from_llm(
11     llm=ChatOpenAI(model="gpt-4", temperature=0.3),
12     retriever=retriever,
13     memory=memory,
14     return_source_documents=True,
15     verbose=True
16 )
17
18 result = qa_chain({"question": query})
```

7.2 Vector Store Configuration

SYNAPSE uses pgvector with PostgreSQL for production vector storage:

```
1 from langchain_community.vectorstores.pgvector import PGVector
2
3 vectorstore = PGVector.from_documents(
4     documents=docs,
5     embedding=embeddings,
6     connection_string=db_url,
7     collection_name="synapse_knowledge",
8     pre_delete_collection=False
9 )
10
11 retriever = vectorstore.as_retriever(
```

```

12     search_type="similarity",
13     search_kwargs={"k": 5}
14 )
15
16 # MMR (Maximal Marginal Relevance) retrieval
17 retriever_mmr = vectorstore.as_retriever(
18     search_type="mmr",
19     search_kwargs={"k": 5, "fetch_k": 20}
20 )

```

7.3 Document Loaders

Multiple document loader implementations support various source formats:

```

1 from langchain_community.document_loaders import (
2     PDFLoader,
3     TextLoader,
4     UnstructuredMarkdownLoader,
5     WebBaseLoader,
6     GitHubRepositoryLoader
7 )
8
9 # PDF documents
10 pdf_loader = PDFLoader("document.pdf")
11 docs = pdf_loader.load()
12
13 # Web content
14 web_loader = WebBaseLoader("https://example.com/docs")
15 web_docs = web_loader.load()
16
17 # GitHub repositories
18 github_loader = GitHubRepositoryLoader(
19     repo_owner="torvalds",
20     repo_name="linux"
21 )

```

7.4 Text Splitters

Documents are split using RecursiveCharacterTextSplitter with configurable chunk parameters:

```

1 from langchain.text_splitter import RecursiveCharacterTextSplitter
2
3 splitter = RecursiveCharacterTextSplitter(
4     chunk_size=1000,
5     chunk_overlap=200,
6     separators=["\n\n", "\n", "\u2013", ""],
7     length_function=len
8 )
9
10 chunks = splitter.split_documents(documents)

```

Configuration Notes:

- Chunk size: 1000 characters (optimal for semantic search)
- Chunk overlap: 200 characters (maintains context continuity)

- Separators: Hierarchical breakdown by paragraph, line, word
- Metadata: Source, page number, chunk index preserved

7.5 Retrieval Strategies

Two primary retrieval strategies are implemented:

7.5.1 Similarity Search

Standard cosine similarity-based retrieval:

```
1 results = vectorstore.similarity_search(  
2     query="machine_learning_frameworks",  
3     k=5  
4 )  
5  
6 for doc in results:  
7     print(f"Score: {doc.metadata.get('score')}")  
8     print(f"Content: {doc.page_content[:200]}")
```

7.5.2 Maximal Marginal Relevance (MMR)

Balances relevance with diversity to avoid redundant results:

```
1 results = vectorstore.max_marginal_relevance_search(  
2     query="python_web_frameworks",  
3     k=5,  
4     fetch_k=20,  
5     lambda_mult=0.5  
6 )
```

7.6 Prompt Templates with Context Injection

RAG chains use specialized prompts that inject retrieved context:

```
1 from langchain.prompts import PromptTemplate  
2  
3 qa_template = """You are a helpful AI assistant.  
4 Answer the following question based on the provided context.  
5 If information is not available in the context, say so.  
6  
7 Context:  
8 {context}  
9  
10 Question: {question}  
11  
12 Provide a detailed answer with citations."""  
13  
14 qa_prompt = PromptTemplate(  
15     template=qa_template,  
16     input_variables=["context", "question"]  
17 )
```

7.7 Source Citation

All RAG responses include source citations for traceability:

```
1 class CitedAnswer(BaseModel):
2     answer: str
3     sources: List[Dict[str, Any]]
4     confidence: float
5
6 def format_with_citations(result):
7     answer = result["answer"]
8     sources = result["source_documents"]
9
10    citations = []
11    for i, doc in enumerate(sources, 1):
12        citations.append({
13            "number": i,
14            "source": doc.metadata.get("source"),
15            "page": doc.metadata.get("page"),
16            "excerpt": doc.page_content[:100]
17        })
18
19    return CitedAnswer(
20        answer=answer,
21        sources=citations,
22        confidence=0.95
23    )
```

Chapter 8

Document Generation Tools

8.1 PDF Generation with ReportLab

The DocumentAgent uses ReportLab 4.x to create professional PDF documents with rich formatting.

8.1.1 PDF Styles and Structure

```
1 from reportlab.lib.pagesizes import letter, A4
2 from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
3 from reportlab.lib.units import inch
4 from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, PageBreak
5 from reportlab.platypus import Table, TableStyle
6 from reportlab.lib import colors
7
8 def create_pdf_report(title, sections, content):
9     doc = SimpleDocTemplate("report.pdf", pagesize=letter)
10    styles = getSampleStyleSheet()
11    story = []
12
13    # Custom styles
14    title_style = ParagraphStyle(
15        'CustomTitle',
16        parent=styles['Heading1'],
17        fontSize=24,
18        textColor=colors.HexColor('#1f77b4'),
19        spaceAfter=30,
20        alignment=1
21    )
22
23    # Title
24    story.append(Paragraph(title, title_style))
25    story.append(Spacer(1, 0.2*inch))
26
27    # Content sections
28    for section in sections:
29        story.append(Paragraph(section['title'], styles['Heading2']))
30        story.append(Paragraph(section['text'], styles['BodyText']))
31        story.append(Spacer(1, 0.1*inch))
32
33    # Build document
34    doc.build(story)
```

8.1.2 Tables and Charts

```

1 from reportlab.graphics.shapes import Drawing
2 from reportlab.graphics.charts.barcharts import VerticalBarChart
3 from reportlab.graphics.charts.linecharts import HorizontalLineChart
4
5 def add_chart_to_pdf(story, chart_data, chart_type='bar'):
6     drawing = Drawing(400, 250)
7
8     if chart_type == 'bar':
9         chart = VerticalBarChart()
10        chart.data = chart_data
11        chart.categoryAxis.labels = ['Q1', 'Q2', 'Q3', 'Q4']
12
13    chart.width = 400
14    chart.height = 250
15    drawing.add(chart)
16    story.append(drawing)
17    story.append(Spacer(1, 0.1*inch))
18
19 def add_table_to_pdf(story, rows, columns):
20     t = Table(rows)
21     t.setStyle(TableStyle([
22         ('BACKGROUND', (0, 0), (-1, 0), colors.grey),
23         ('TEXTCOLOR', (0, 0), (-1, 0), colors.whitesmoke),
24         ('ALIGN', (0, 0), (-1, -1), 'CENTER'),
25         ('FONTNAME', (0, 0), (-1, 0), 'Helvetica-Bold'),
26         ('FONTSIZE', (0, 0), (-1, 0), 14),
27         ('BOTTOMPADDING', (0, 0), (-1, 0), 12),
28         ('BACKGROUND', (0, 1), (-1, -1), colors.beige),
29         ('GRID', (0, 0), (-1, -1), 1, colors.black)
30     ]))
31     story.append(t)
32     story.append(Spacer(1, 0.1*inch))

```

8.2 PowerPoint Generation with python-pptx

DocumentAgent uses python-pptx 0.6.x to create dynamic presentations.

8.2.1 Slide Layouts and Content

```

1 from pptx import Presentation
2 from pptx.util import Inches, Pt
3 from pptx.enum.text import PP_ALIGN
4 from pptx.dml.color import RGBColor
5
6 def create_presentation(title, slides):
7     prs = Presentation()
8     prs.slide_width = Inches(10)
9     prs.slide_height = Inches(7.5)
10
11    # Title slide
12    title_slide_layout = prs.slide_layouts[0]
13    slide = prs.slides.add_slide(title_slide_layout)
14    title = slide.shapes.title
15    subtitle = slide.placeholders[1]

```

```

16
17 title.text = "Quarterly_Technology_Report"
18 subtitle.text = "Q4_2024"
19
20 # Content slides
21 for slide_config in slides:
22     blank_slide_layout = prs.slide_layouts[6]
23     slide = prs.slides.add_slide(blank_slide_layout)
24
25     # Add title
26     left = Inches(0.5)
27     top = Inches(0.5)
28     width = Inches(9)
29     height = Inches(1)
30
31     title_box = slide.shapes.add_textbox(left, top, width, height)
32     title_frame = title_box.text_frame
33     title_frame.word_wrap = True
34     p = title_frame.paragraphs[0]
35     p.text = slide_config['title']
36     p.font.size = Pt(44)
37     p.font.bold = True
38
39     # Add content
40     content_box = slide.shapes.add_textbox(
41         Inches(0.5), Inches(1.5), Inches(9), Inches(5.5)
42     )
43     tf = content_box.text_frame
44     tf.word_wrap = True
45
46     for bullet_text in slide_config.get('bullets', []):
47         p = tf.add_paragraph()
48         p.text = bullet_text
49         p.level = 0
50
51 prs.save('presentation.pptx')

```

8.3 Word Document Generation with python-docx

DocumentAgent uses python-docx 0.8.x for Word document creation.

```

1 from docx import Document
2 from docx.shared import Inches, Pt, RGBColor
3 from docx.enum.text import WD_ALIGN_PARAGRAPH
4
5 def create_word_document(title, content, include_toc=True):
6     doc = Document()
7
8     # Title
9     title_para = doc.add_paragraph(title)
10    title_para.style = 'Heading_1'
11    title_para.alignment = WD_ALIGN_PARAGRAPH.CENTER
12
13    # Table of Contents (if enabled)
14    if include_toc:
15        doc.add_paragraph('Table_of_Contents', style='Heading_2')
16        # Word will auto-generate TOC
17

```

```
18 # Content sections
19 for section in content['sections']:
20     heading = doc.add_paragraph(section['title'], style='Heading_2')
21     doc.add_paragraph(section['text'])
22
23     if 'bullets' in section:
24         for bullet in section['bullets']:
25             doc.add_paragraph(bullet, style='List_Bullet')
26
27     doc.add_paragraph() # Spacing
28
29 doc.save('document.docx')
```

Chapter 9

Project Builder Tool

9.1 Project Template Architecture

The ProjectAgent generates complete project structures for multiple frameworks.

9.1.1 Django REST API Template

```
1 DJANGO_REST_STRUCTURE = {
2     "project_name": "myproject",
3     "app_name": "myapp",
4     "files": {
5         "manage.py": "#_Django_management_script",
6         "requirements.txt": "Django==4.2\nDjangoRESTframework==3.14\npsycopg2-binary==2.9",
7         "myproject/settings.py": "#_Django_settings",
8         "myproject/urls.py": "#_URL_routing",
9         "myapp/models.py": "from_django.db_import_models\nnclass_BaseModel(models.Model):_pass",
10        "myapp/serializers.py": "from_rest_framework_import_serializers",
11        "myapp/views.py": "from_rest_framework_import_viewsets",
12        "myapp/urls.py": "#_App_URLs"
13    }
14 }
```

9.1.2 FastAPI Microservice Template

```
1 FASTAPI_STRUCTURE = {
2     "project_name": "my_service",
3     "files": {
4         "main.py": """from fastapi import FastAPI
5
6 app = FastAPI()
7
8 @app.get("/health")
9 async def health_check():
10     return {"status": "ok"}
11 """,
12     "requirements.txt": "fastapi==0.104\nuvicorn==0.24",
13     "middleware.py": "#_Custom_middleware",
14     "routers/users.py": "from_fastapi_import_APIRouter\nrouter=_APIRouter()",
15     ".env": "DATABASE_URL=postgresql://user:pass@localhost/db",
16     "docker-compose.yml": "version:_3.8\nservices:\n    _api:\n        build:_."
17 }
```

```
18 }
```

9.1.3 Next.js Application Template

```
1 NEXTJS_STRUCTURE = {
2   "project_name": "my-app",
3   "files": {
4     "package.json": ""{"name": "my-app", "version": "1.0.0",
5 "dependencies": {"next": "^14.0", "react": "^18.0"}}"",
6     "next.config.js": "module.exports={}",
7     "app/page.js": ""export default function Home() {
8 return <main>Welcome</main>
9 }"",
10    "app/layout.js": ""export const metadata = {title: 'App'}
11 export default function RootLayout({children}) {
12 return <html>{children}</html>
13 }"",
14    "tailwind.config.js": "module.exports={theme:{}}",
15    ".gitignore": "node_modules\n.next\n.env.local"
16  }
17 }
```

9.1.4 React Component Library Template

```
1 REACT_LIB_STRUCTURE = {
2   "project_name": "my-components",
3   "files": {
4     "package.json": ""{"name": "my-components", "type": "module",
5 "exports": {". /Button": ". /src/Button.jsx"}}"",
6     "src/Button.jsx": ""export default function Button({children}) {
7 return <button>{children}</button>
8 }"",
9     "src/index.js": "export {default as Button} from './Button'",
10    "tsconfig.json": "{ \"compilerOptions\": { \"jsx\": \"react-jsx\" } }",
11    "vite.config.js": "import react from '@vitejs/plugin-react'"
12  }
13 }
```

9.1.5 Python Data Science Template

```
1 DATASCIENCE_STRUCTURE = {
2   "project_name": "my_analysis",
3   "files": {
4     "requirements.txt": "pandas==2.0\nnumpy==1.24\nscikit-learn==1.3\njupyter==1.0",
5     "notebooks/01_exploration.ipynb": "#_Jupyter_notebook",
6     "src/data_loader.py": ""import pandas as pd
7
8 def load_data(path):
9     return pd.read_csv(path)
10    "",
11    "src/model.py": ""from sklearn.ensemble import RandomForestClassifier
12
13 class Model:
14     def __init__(self):
```

```

15         self.model = RandomForestClassifier()
16     """
17     "README.md": "#_Data_Science_Project",
18     ".gitignore": "__pycache__\n*.ipynb_checkpoints\n.env"
19 }
20 }

```

9.2 Project Generation Implementation

```

1  import os
2  from pathlib import Path
3
4  def create_project(project_type, name, features=None):
5      template = PROJECT_TEMPLATES[project_type]
6      project_path = Path(name)
7      project_path.mkdir(exist_ok=True)
8
9      files_created = 0
10
11     # Create file structure
12     for file_path, content in template['files'].items():
13         full_path = project_path / file_path
14         full_path.parent.mkdir(parents=True, exist_ok=True)
15
16         with open(full_path, 'w') as f:
17             f.write(content)
18
19         files_created += 1
20
21     # Add optional features
22     if features:
23         if 'auth' in features:
24             add_authentication(project_path, project_type)
25         if 'testing' in features:
26             add_testing(project_path, project_type)
27         if 'ci_cd' in features:
28             add_ci_cd(project_path, project_type)
29
30     # Initialize git
31     os.chdir(project_path)
32     os.system('git_init')
33
34     return {
35         "project_path": str(project_path.absolute()),
36         "files_created": files_created,
37         "next_steps": [
38             f"cd_{name}",
39             "Install_dependencies",
40             "Configure_environment_variables",
41             "Run_development_server"
42         ]
43     }

```

Chapter 10

Task Queue Integration

10.1 Celery Configuration

SYNAPSE uses Celery for asynchronous agent task execution with Redis/PostgreSQL backend.

```
1 from celery import Celery, Task
2 from kombu import Queue
3
4 app = Celery('synapse')
5 app.conf.update(
6     broker_url='redis://localhost:6379',
7     result_backend='db+postgresql://user:pass@localhost/synapse',
8     task_serializer='json',
9     accept_content=['json'],
10    result_expires=3600,
11    task_track_started=True,
12    task_time_limit=600,
13    worker_prefetch_multiplier=1
14 )
15
16 app.conf.task_queues = (
17     Queue('research_tasks'),
18     Queue('document_tasks'),
19     Queue('trend_tasks'),
20     Queue('project_tasks'),
21 )
```

10.2 Agent Execution as Celery Task

```
1 @app.task(bind=True, queue='research_tasks')
2 def execute_research_agent(self, user_id, task_description):
3     agent = ResearchAgent()
4
5     try:
6         self.update_state(state='PROGRESS', meta={'current': 0, 'total': 10})
7
8         result = agent.execute(
9             task=task_description,
10            context={"user_id": user_id},
11            callback=self.update_progress
12        )
```

```

13
14     return {
15         "status": "completed",
16         "result": result,
17         "user_id": user_id,
18         "timestamp": datetime.now().isoformat()
19     }
20
21 except Exception as e:
22     self.update_state(state='FAILURE', meta={'error': str(e)})
23     raise
24
25 def update_progress(self, current, total, message=""):
26     self.update_state(state='PROGRESS',
27         meta={'current': current, 'total': total, 'message': message})

```

10.3 Task Status Tracking

```

1 from celery.result import AsyncResult
2
3 class TaskTracker:
4     def __init__(self, task_id):
5         self.task_id = task_id
6         self.result = AsyncResult(task_id)
7
8     def get_status(self):
9         return {
10             "task_id": self.task_id,
11             "state": self.result.state,
12             "progress": self.result.info.get('current', 0),
13             "total": self.result.info.get('total', 0),
14             "ready": self.result.ready(),
15             "successful": self.result.successful(),
16             "failed": self.result.failed()
17         }
18
19     def get_result(self):
20         if self.result.successful():
21             return self.result.result
22         elif self.result.failed():
23             return {"error": str(self.result.info)}
24         else:
25             return {"status": self.result.state}

```

Chapter 11

Agent Safety and Constraints

11.1 Tool Use Validation

All tool calls are validated before execution to prevent misuse.

```
1 class ToolValidator:
2     def __init__(self):
3         self.dangerous_tools = [
4             'delete_database',
5             'modify_system_config',
6             'access_production'
7         ]
8
9     def validate_tool_call(self, tool_name, tool_input, user_tier):
10        # Check dangerous tools
11        if tool_name in self.dangerous_tools:
12            if user_tier != 'admin':
13                raise PermissionError(f"User_cannot_call_{tool_name}")
14
15        # Validate input parameters
16        if not self.validate_schema(tool_input):
17            raise ValueError("Invalid_tool_input_schema")
18
19        # Check resource limits
20        if not self.check_resource_limits(tool_name, user_tier):
21            raise ResourceError("Resource_limit_exceeded")
22
23        return True
```

11.2 Output Sanitization

Agent outputs are sanitized to remove sensitive information.

```
1 import re
2
3 class OutputSanitizer:
4     SENSITIVE_PATTERNS = [
5         r'(password|api_key|secret) ["\']?\s*[:=]\s*["\']?([^"\'\s]+)',
6         r'(token|Bearer)\s+([A-Za-z0-9\-. _]+)',
7         r'(email|email_address)\. *?([A-Za-z0-9\-. _%+~]+@[A-Za-z0-9\-. _]+)',
8         r'(phone|telephone) ["\']?\s*[:=]\s*([0-9\-\+\(\)\s]+)'
9     ]
```

```

10
11 @staticmethod
12 def sanitize(text):
13     sanitized = text
14     for pattern in OutputSanitizer.SENSITIVE_PATTERNS:
15         sanitized = re.sub(pattern, r'\1: [REDACTED]', sanitized, flags=re.IGNORECASE)
16     return sanitized

```

11.3 Cost Limits and Timeout Handling

```

1 class ExecutionLimiter:
2     def __init__(self, max_tokens_per_task=10000, timeout_per_tool=30,
3                 timeout_per_run=300):
4         self.max_tokens = max_tokens_per_task
5         self.tool_timeout = timeout_per_tool
6         self.run_timeout = timeout_per_run
7         self.token_counter = CostTracker()
8
9     def check_token_budget(self, tokens_used):
10        if tokens_used > self.max_tokens:
11            raise TokenLimitExceeded(f"Exceeded_{self.max_tokens}_token_limit")
12        return True
13
14    def enforce_tool_timeout(self, tool_name, func, args):
15        import signal
16
17        def timeout_handler(signum, frame):
18            raise TimeoutError(f"Tool_{tool_name}_exceeded_{self.tool_timeout}s")
19
20        signal.signal(signal.SIGALRM, timeout_handler)
21        signal.alarm(self.tool_timeout)
22
23        try:
24            result = func(*args)
25            signal.alarm(0)
26            return result
27        except Exception as e:
28            signal.alarm(0)
29            raise

```

11.4 Rate Limiting

```

1 from datetime import datetime, timedelta
2
3 class RateLimiter:
4     def __init__(self):
5         self.user_limits = {
6             'free': {'requests_per_hour': 10, 'tokens_per_day': 5000},
7             'pro': {'requests_per_hour': 100, 'tokens_per_day': 100000},
8             'enterprise': {'requests_per_hour': 1000, 'tokens_per_day': 1000000}
9         }
10        self.request_tracker = {}
11
12    def is_allowed(self, user_id, user_tier):

```

```

13     now = datetime.now()
14     hour_ago = now - timedelta(hours=1)
15
16     if user_id not in self.request_tracker:
17         self.request_tracker[user_id] = []
18
19     # Clean old requests
20     self.request_tracker[user_id] = [
21         req_time for req_time in self.request_tracker[user_id]
22         if req_time > hour_ago
23     ]
24
25     limit = self.user_limits[user_tier]['requests_per_hour']
26
27     if len(self.request_tracker[user_id]) >= limit:
28         return False
29
30     self.request_tracker[user_id].append(now)
31     return True

```

11.5 Dangerous Action Confirmation

```

1 class ConfirmationRequester:
2     DANGEROUS_ACTIONS = {
3         'delete_project': 'permanent_deletion',
4         'reset_database': 'complete_data_reset',
5         'modify_access_control': 'security_settings_change',
6         'deploy_to_production': 'production_deployment'
7     }
8
9     @staticmethod
10    def requires_confirmation(action_name):
11        return action_name in ConfirmationRequester.DANGEROUS_ACTIONS
12
13    @staticmethod
14    def get_confirmation_prompt(action_name):
15        risk = ConfirmationRequester.DANGEROUS_ACTIONS.get(action_name)
16        return f"This {risk} will result in {risk}. Please confirm (yes/no): "

```

Chapter 12

Open Source Libraries

SYNAPSE leverages industry-standard open source libraries for AI agent implementation.

12.1 Core Dependencies

The primary dependencies for SYNAPSE agent layer:

- **LangChain 0.1.x:** Agent orchestration and LLM interactions
- **langchain-openai:** OpenAI LLM integration
- **langchain-community:** Community tools and integrations
- **langgraph:** Complex workflow and agent graph management
- **ReportLab 4.x:** PDF document generation
- **python-pptx 0.6.x:** PowerPoint creation
- **python-docx 0.8.x:** Word document generation
- **chromadb:** Alternative vector store implementation
- **tiktoken:** OpenAI token counting and encoding

12.2 Installation

```
1 pip install langchain==0.1.0
2 pip install langchain-openai==0.0.5
3 pip install langchain-community==0.0.10
4 pip install langgraph==0.0.8
5 pip install reportlab==4.0.9
6 pip install python-pptx==0.6.21
7 pip install python-docx==0.8.11
8 pip install chromadb==0.4.21
9 pip install tiktoken==0.5.2
```

Chapter 13

Agent Monitoring and Observability

13.1 Execution Logging

All agent thoughts, actions, and observations are logged for transparency.

```
1 import logging
2 from datetime import datetime
3
4 logger = logging.getLogger('synapse.agents')
5 logger.setLevel(logging.DEBUG)
6
7 class AgentExecutionLogger:
8     @staticmethod
9     def log_thought(agent_id, iteration, thought):
10         logger.info(f"[Agent_{agent_id},Iter_{iteration}]_THOUGHT:_{thought}")
11
12     @staticmethod
13     def log_action(agent_id, iteration, tool_name, tool_input):
14         logger.info(f"[Agent_{agent_id},Iter_{iteration}]_ACTION:_{tool_name}")
15         logger.debug(f"Input:_{tool_input}")
16
17     @staticmethod
18     def log_observation(agent_id, iteration, observation):
19         logger.info(f"[Agent_{agent_id},Iter_{iteration}]_OBSERVATION:_{observation[:200]}")
20
21     @staticmethod
22     def log_final_answer(agent_id, answer):
23         logger.info(f"[Agent_{agent_id}]_FINAL_ANSWER:_{answer[:200]}")
```

13.2 Cost Tracking

Detailed cost tracking per agent run and per user.

```
1 class CostAnalytics:
2     def __init__(self, db_connection):
3         self.db = db_connection
4
5     def log_execution_cost(self, agent_id, user_id, input_tokens,
6                           output_tokens, duration_seconds):
7         cost = self.calculate_cost(input_tokens, output_tokens)
8
9         record = {
```

```

10         'agent_id': agent_id,
11         'user_id': user_id,
12         'input_tokens': input_tokens,
13         'output_tokens': output_tokens,
14         'total_cost_usd': cost,
15         'duration_seconds': duration_seconds,
16         'timestamp': datetime.now()
17     }
18
19     self.db.insert('execution_costs', record)
20
21     @staticmethod
22     def calculate_cost(input_tokens, output_tokens):
23         input_cost = (input_tokens / 1000) * 0.03
24         output_cost = (output_tokens / 1000) * 0.06
25         return input_cost + output_cost

```

13.3 Success and Failure Metrics

```

1 class MetricsCollector:
2     def __init__(self, db_connection):
3         self.db = db_connection
4
5     def record_execution(self, agent_type, task, success, duration,
6                         error_message=None):
7         metric = {
8             'agent_type': agent_type,
9             'task_type': task,
10            'success': success,
11            'duration_seconds': duration,
12            'error_message': error_message,
13            'timestamp': datetime.now()
14        }
15
16        self.db.insert('execution_metrics', metric)
17
18    def get_agent_success_rate(self, agent_type, period_days=30):
19        query = f"""
20        SELECT
21            COUNT(*) as total,
22            SUM(CASE WHEN success THEN 1 ELSE 0 END) as successful
23        FROM execution_metrics
24        WHERE agent_type = %s
25        AND timestamp > NOW() - INTERVAL '{period_days} days'
26        """
27
28        result = self.db.execute(query, [agent_type])
29        total, successful = result[0]
30        return successful / total if total > 0 else 0

```

13.4 Tool Usage Statistics

```

1 class ToolUsageTracker:
2     def __init__(self, db_connection):

```

```

3         self.db = db_connection
4
5     def record_tool_usage(self, tool_name, agent_id, success,
6                           duration_seconds, error=None):
7         record = {
8             'tool_name': tool_name,
9             'agent_id': agent_id,
10            'success': success,
11            'duration_seconds': duration_seconds,
12            'error': error,
13            'timestamp': datetime.now()
14        }
15
16        self.db.insert('tool_usage', record)
17
18    def get_tool_statistics(self, tool_name, period_days=30):
19        query = """
20        SELECT
21            tool_name,
22            COUNT(*) as total_calls,
23            SUM(CASE WHEN success THEN 1 ELSE 0 END) as successful_calls,
24            AVG(duration_seconds) as avg_duration,
25            MAX(duration_seconds) as max_duration
26        FROM tool_usage
27        WHERE tool_name = %s
28        AND timestamp > NOW() - INTERVAL '%s days'
29        GROUP BY tool_name
30        """
31
32        return self.db.execute(query, [tool_name, period_days])

```

13.5 LangSmith Integration

Integration with LangSmith for comprehensive agent trace visualization and debugging.

```

1 from langsmith import traceable
2 from langsmith.client import Client
3
4 client = Client()
5
6 @traceable(
7     run_type="chain",
8     tags=["synapse", "agent"]
9 )
10 def execute_with_tracing(agent, input_data):
11     return agent.invoke(input_data)
12
13 class LangSmithIntegration:
14     def __init__(self, api_key):
15         self.client = Client(api_key=api_key)
16
17     def get_execution_trace(self, run_id):
18         run = self.client.read_run(run_id)
19         return {
20             "run_id": run.id,
21             "status": run.status,
22             "duration": (run.end_time - run.start_time).total_seconds(),
23             "inputs": run.inputs,

```

```
24         "outputs": run.outputs,
25         "child_runs": [child.id for child in run.child_runs]
26     }
27
28     def export_trace_data(self, run_id, format='json'):
29         trace = self.get_execution_trace(run_id)
30
31         if format == 'json':
32             import json
33             return json.dumps(trace, indent=2)
34
35         return trace
```

Chapter 14

Conclusion

The SYNAPSE AI Agent layer represents a comprehensive system for autonomous task execution using LangChain ReAct agents. This specification covers all critical components including agent architecture, specialized agent types, tool definitions, memory systems, RAG integration, document generation, project scaffolding, safety mechanisms, and monitoring capabilities.

Key design principles emphasize transparency through detailed logging, cost tracking and budgeting, safety through validation and rate limiting, and extensibility through modular tool architecture. The system supports complex multi-step reasoning while maintaining user control and visibility into agent decision-making.